

NMAP

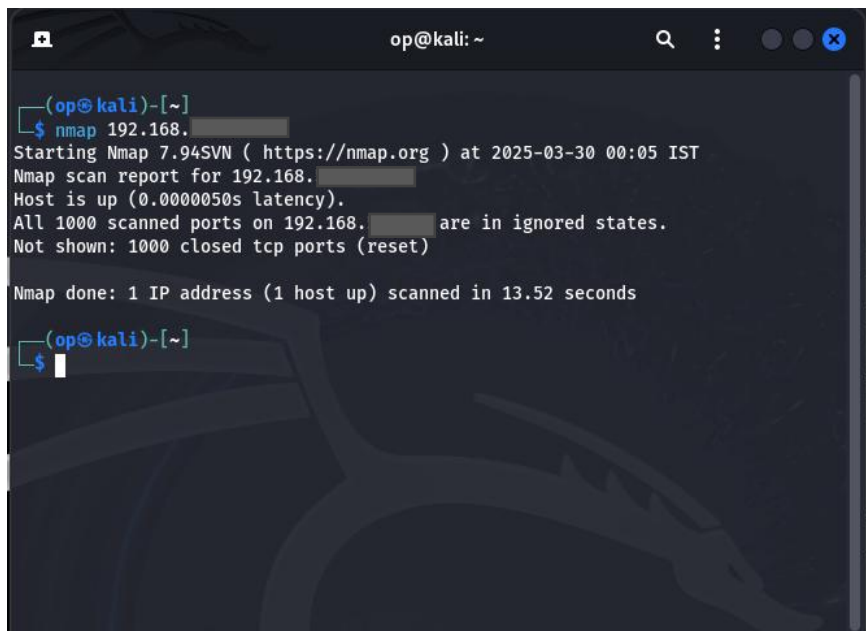
- **Introduction**

[Network security is a critical aspect of cybersecurity. Nmap \(Network Mapper\) is a powerful open-source tool used for network discovery and security auditing. In this task, we performed different types of Nmap scans to analyze a local network, identify open ports, and detect running services.](#)

- **Network Scanning with Nmap**

Perform different types of scans:

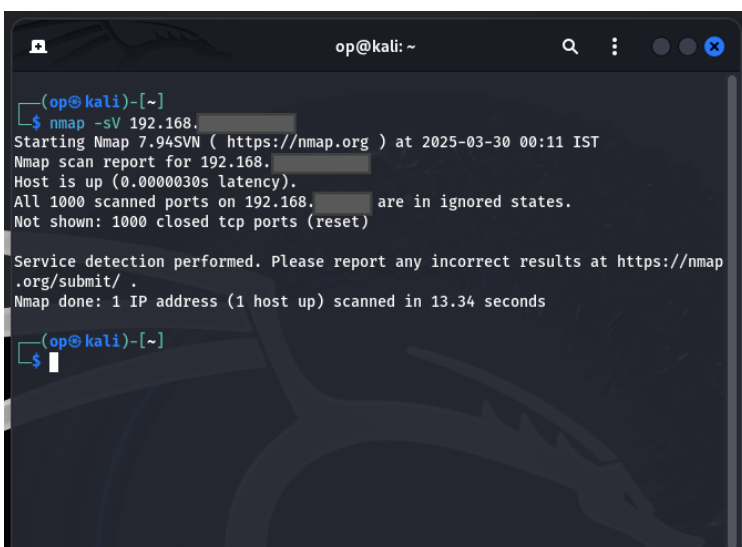
1. Basic scan: nmap <IP>



```
op@kali: ~  
(op@kali)-[~]  
$ nmap 192.168.1.1  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-03-30 00:05 IST  
Nmap scan report for 192.168.1.1  
Host is up (0.0000050s latency).  
All 1000 scanned ports on 192.168.1.1 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Nmap done: 1 IP address (1 host up) scanned in 13.52 seconds  
  
(op@kali)-[~]  
$
```

[This shows open ports and general host information.](#)

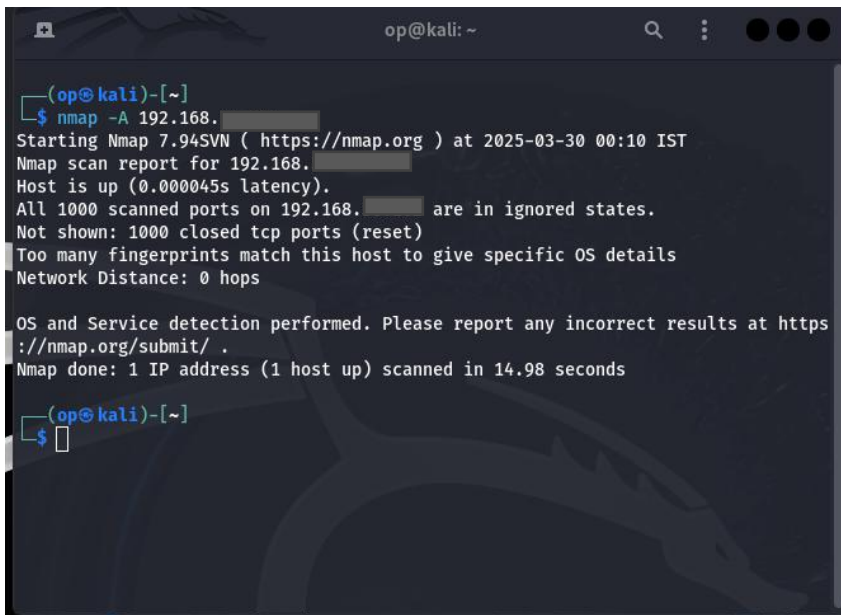
2. Service detection: nmap -sV <IP>



```
op@kali: ~  
(op@kali)-[~]  
$ nmap -sV 192.168.1.1  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-03-30 00:11 IST  
Nmap scan report for 192.168.1.1  
Host is up (0.0000030s latency).  
All 1000 scanned ports on 192.168.1.1 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 13.34 seconds  
  
(op@kali)-[~]  
$
```

[This detects which services are running on open ports.](#)

3. Aggressive scan: nmap -A <IP>

A terminal window titled 'op@kali: ~' showing the execution of an aggressive Nmap scan. The command 'nmap -A 192.168.1.100' is entered. The output indicates the scan was performed on 2025-03-30 at 00:10 IST. It reports that the host is up with a latency of 0.000045s. All 1000 scanned ports are in ignored states, and 1000 closed TCP ports were not shown. The network distance is 0 hops. OS and service detection were performed, and the scan was completed in 14.98 seconds.

```
(op@kali)-[~]
$ nmap -A 192.168.1.100
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-03-30 00:10 IST
Nmap scan report for 192.168.1.100
Host is up (0.000045s latency).
All 1000 scanned ports on 192.168.1.100 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.98 seconds

(op@kali)-[~]
$
```

This scan provides:

- OS detection
- Service versions
- Scripts for vulnerability detection